Example 1:

The page of a level book is given below. Calculate the RL of all the points by any one of the methods. Apply usual checks. Assume the RL of BM as 25.000m.

Station	BS	IS	FS	Remarks
A	1.120			BM
B		1.650		
C		1.750		
D	1.200		2.900	CP1
E		1.660		
F		2.520		
G	0.900		3.250	CP2
H	1560		2.100	
I			2.200	

Solution

Type (I) Height of collimation method

The observation are tabulated and reduced levels are obtained as follows.

Station	BS	IS	FS	HC or HI	RL	Remarks
A	1.120			26.120	25.000	BM
B		1.650			24.470	
C		1.750			24.370	
D	1.200		2.900	24.420	23.220	CP1
E		1.660			22.760	
F		2.520			21.900	
G	0.900		3.250	22.070	21.170	CP2
H	1.560		2.100	21.530	19.970	
I			2.200		19.330	
Total	4.780		10.450			

Calculations

HC for I set up = RL of A + BS of A = 25.000 + 1.120 = 26.120

RL of B = HC for I set up – IS of B = 26.120 - 1.650 = 24.470

RL of C = HC for I set up – IS of C = 26.120 - 1.750 = 24.370

RL of D = HC for I set up – IS of D = 26.120 - 2.900 = 23.220

HC for II set up = RL of D + BS of D = 23.220 + 1.200 = 24.420

RL of E = HC for II set up – IS of E = 24.420 - 1.660 = 22.760

RL of F = HC for II set up – IS of F = 24.420 - 2.520 = 21.900

RL of G = HC for II set up – IS of G = 24.420 - 3.250 = 21.170

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